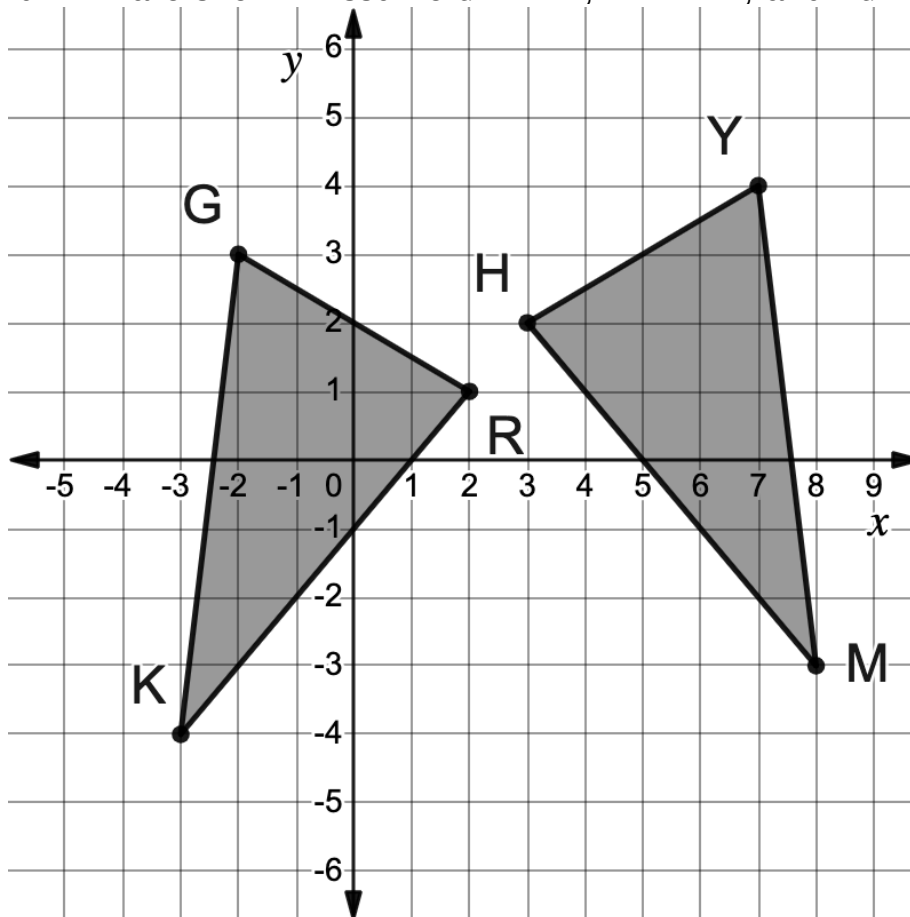


Triangles GRK and MHY are shown. Assume $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, and $\overline{KG} \cong \overline{YM}$.



1. Explain how to use geometric tools to verify that $\triangle HYM \cong \triangle RGK$.
2. Write a sequence of transformations that will map $\triangle HYM$ onto $\triangle RGK$.

Show your work for questions 3 through 5 on the coordinate grid.

3. Draw $\triangle H'Y'M'$ the result of the first transformation.
4. Map the vertices of $\triangle HYM$ to $\triangle H'Y'M'$.
5. Map the vertices of $\triangle H'Y'M'$ to $\triangle RGK$.

For questions 6 through 8, complete the statements.

6. The outcome of mapping the transformations of $\triangle HYM$ to $\triangle H'Y'M'$ and then from $\triangle H'Y'M'$ to $\triangle$ _____ shows that point H' coincides with point _____, point Y' coincides with point _____, and point M' coincides with point _____.

7. Because the definition of _____ in terms of _____
 _____ preserves _____,
 it can be stated that $\triangle HYM \cong \triangle$ _____.

8. Since it was assumed that $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, and $\overline{KG} \cong \overline{YM}$ then _____ is sufficient to show the triangles are congruent.

9. Select all of the given statements that would result in a sequence of transformations that would show SAS congruence is sufficient to show the triangles are congruent.

- ☐ $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, $\angle GRK \cong \angle YHM$
- ☐ $\overline{GR} \cong \overline{HY}$, $\overline{KG} \cong \overline{YM}$, $\angle GRK \cong \angle YHM$
- ☐ $\overline{KG} \cong \overline{YM}$, $\overline{RK} \cong \overline{HM}$, $\angle RKG \cong \angle YMH$
- ☐ $\overline{KG} \cong \overline{YM}$, $\overline{RK} \cong \overline{HM}$, $\angle RGK \cong \angle MYH$
- ☐ $\overline{RK} \cong \overline{HM}$, $\overline{GR} \cong \overline{HY}$, $\angle RGK \cong \angle MYH$
- ☐ $\overline{RK} \cong \overline{HM}$, $\overline{KG} \cong \overline{YM}$, $\angle GRK \cong \angle YHM$

10. Select all of the given statements that would result in a sequence of transformations that would show ASA congruence is sufficient to show the triangles are congruent.

- ☐ $\angle RGK \cong \angle MYH$, $\angle GRK \cong \angle YHM$, $\overline{RK} \cong \overline{HM}$
- ☐ $\angle RGK \cong \angle MYH$, $\angle GRK \cong \angle YHM$, $\overline{GR} \cong \overline{HY}$
- ☐ $\angle GRK \cong \angle YHM$, $\angle RGK \cong \angle MYH$, $\overline{KG} \cong \overline{YM}$
- ☐ $\angle RKG \cong \angle YMH$, $\angle RGK \cong \angle MYH$, $\overline{KG} \cong \overline{YM}$
- ☐ $\angle RKG \cong \angle YMH$, $\angle GRK \cong \angle YHM$, $\overline{RK} \cong \overline{HM}$
- ☐ $\angle RKG \cong \angle YMH$, $\angle RGK \cong \angle MYH$, $\overline{GR} \cong \overline{HY}$